CSE422: Artificial intelligence

Naïve-Bayes Classifier

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BAYES' Rule

- From the formula of conditional probability P(x | y) = P(x, y) / P(y), we can infer: P(x, y) = P (x | y) × P(y) [called the product rule]
- Similarly, P(x, y) = P (y | x) × P(x)
- Dividing we get:

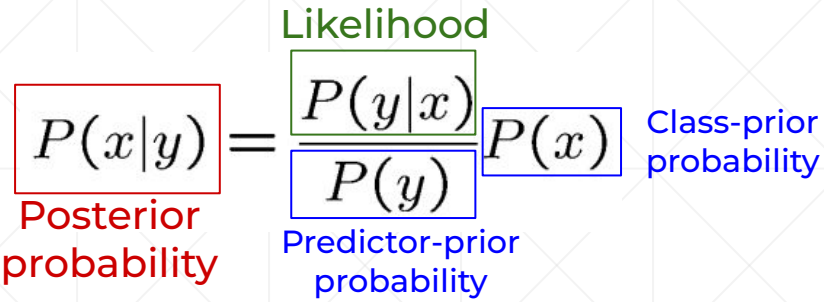
$$P(x|y) = \frac{P(y|x)}{P(y)} P(x)$$

- In other words:

$$P(\text{cause}|\text{effect}) = \frac{P(\text{effect}|\text{cause})P(\text{cause})}{P(\text{effect})}$$

- Why is this SO important?
  - It is hard to know the cause, but it is easy to see the effect
  - Using this formula, we can infer the cause by analyzing the effect

BAYES' Rule


Likelihood
$P(x|y) = \frac{P(y|x)}{P(y)} P(x)$
Class-prior probability
Posterior probability
Predictor-prior probability

BAYES' Rule

Temperature (F)	Play Tennis
70	Yes
32	No
65	No
75	Yes
30	No
75	Yes
72	No

- Consider two boolean RVs, A: "We will play tennis" and B: "Warm day (Temp >= 50 F)"
  - Think ML: Temperature is ***"feature"***, Play tennis is ***"target"***
- If it is a warm day, what is the probability that we will play tennis?
  - Think ML: Given a ***feature***, predict the ***target***

BAYES' Rule

Temperature (F)	Play Tennis
70	Yes
32	No
65	No
75	Yes
30	No
75	Yes
72	No

- If it is a warm day, what is the probability that we will play tennis?
- P(b | a) = 1
- P(a) = 3/7
- P(b) = 5/7
- Therefore, P(a | b) = 0.6
  - Think ML: **Output = YES**

Why BAYES is DIFFICULT in Practice

Wind	Humidity	Temperature	Play Tennis
5	95	70	Yes
10	80	32	No
20	80	65	No
10	85	75	Yes
8	35	30	No
8	35	75	Yes
25	35	72	No

- Now let there are FOUR boolean RVs: A (“Play tennis”), B(“Warm day”), C(“Dry day”), D(“Windy day”) [Think ML: Features: B, C, D, Target: A]
- We want to find the probability of playing tennis in a warm, dry, windy day P(a | b, c, d)
- For this we need the likelihood P(b, c, d | a)
- Computing this is REALLY difficult and makes it hard to use BAYES in real applications
  - For this problem it is actually easy (do it)
  - But this is a small table with 7 entries only
  - In real world, to compute such a likelihood with good accuracy would require an extensive record with millions of entries – very expensive and difficult

Naïve-Bayes Classifier

- Computing the likelihood P(b, c, d | a) is difficult in real world applications
- However, we can APPROXIMATE the calculation, making it practical for real use
- The **Naïve assumption**: All **features** are **conditionally independent** of each other given the target label
- In other words, B, C, D are conditionally independent given A
- So, using our formula for conditional independence, we get:
- $P(b, c, d \mid a) = P(b \mid a) \times P(c \mid a) \times P(d \mid a)$
- This is much easier to calculate from records!
- The Naïve Bayes classifier often does surprisingly well, outperforming more sophisticated classification methods
- $P(X1, X2, \cdots, Xn \mid C) = P(X1 \mid C)P(X2 \mid C) \cdots P(Xn \mid C)$

Approximation using Naïve-BAYES

Wind	Humidity	Temperature	Play Tennis
5	95	70	Yes
10	80	32	No
20	80	65	No
10	85	75	Yes
8	35	30	No
8	35	75	Yes
25	35	72	No

- We want to find the probability of playing tennis in a warm, dry, windy day P(a | b, c, d)
- P(a | b, c, d) = P(b, c, d | a) × P (a) / P(b, c, d)
- Do it yourself!

Naïve-BAYES Example

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Step 1 – **Learning:** From the training data, we will **LEARN the conditional probability** distribution of each **Feature** (Outlook, Temperature, Humidity, Wind) Given the **Target** (PlayTennis)

Naïve-BAYES Example

PlayTennis: training examples

Day	Outlook	Temperature	Humidity	Wind	PlayTennis
D1	Sunny	Hot	High	Weak	No
D2	Sunny	Hot	High	Strong	No
D3	Overcast	Hot	High	Weak	Yes
D4	Rain	Mild	High	Weak	Yes
D5	Rain	Cool	Normal	Weak	Yes
D6	Rain	Cool	Normal	Strong	No
D7	Overcast	Cool	Normal	Strong	Yes
D8	Sunny	Mild	High	Weak	No
D9	Sunny	Cool	Normal	Weak	Yes
D10	Rain	Mild	Normal	Weak	Yes
D11	Sunny	Mild	Normal	Strong	Yes
D12	Overcast	Mild	High	Strong	Yes
D13	Overcast	Hot	Normal	Weak	Yes
D14	Rain	Mild	High	Strong	No

Learning Phase

Outlook	Play=*Yes*	Play=*No*
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=*Yes*	Play=*No*
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Play=*Yes*	Play=*No*
High	3/9	4/5
Normal	6/9	1/5

Wind	Play=*Yes*	Play=*No*
Strong	3/9	3/5
Weak	6/9	2/5

$P(\text{Play}=Yes) = 9/14$

$P(\text{Play}=No) = 5/14$

Caution: This is a **CONDITIONAL** probability table, NOT Joint Probability Distribution Table like toothache, cavity tables

Naïve-BAYES Example

Step 2 – **Testing:** Using what we learned (likelihoods) during training, we will **PREDICT** the label (target) for a new, unseen input data

Learning Phase

Outlook	Play=*Yes*	Play=*No*
Sunny	2/9	3/5
Overcast	4/9	0/5
Rain	3/9	2/5

Temperature	Play=*Yes*	Play=*No*
Hot	2/9	2/5
Mild	4/9	2/5
Cool	3/9	1/5

Humidity	Play=*Yes*	Play=*No*
High	3/9	4/5
Normal	6/9	1/5

Wind	Play=*Yes*	Play=*No*
Strong	3/9	3/5
Weak	6/9	2/5

$P(\text{Play}=Yes) = 9/14$

$P(\text{Play}=No) = 5/14$

Test Phase

- Given a new instance, predict its label

  $\mathbf{x}'$=(Outlook=*Sunny*, Temperature=*Cool*, Humidity=*High*, Wind=*Strong*)

- Look up tables achieved in the learning phrase

  $P(\text{Outlook}=Sunny \mid \text{Play}=Yes) = 2/9$
  $P(\text{Temperature}=Cool \mid \text{Play}=Yes) = 3/9$
  $P(\text{Huminity}=High \mid \text{Play}=Yes) = 3/9$
  $P(\text{Wind}=Strong \mid \text{Play}=Yes) = 3/9$
  $P(\text{Play}=Yes) = 9/14$

  $P(\text{Outlook}=Sunny \mid \text{Play}=No) = 3/5$
  $P(\text{Temperature}=Cool \mid \text{Play}==No) = 1/5$
  $P(\text{Huminity}=High \mid \text{Play}=No) = 4/5$
  $P(\text{Wind}=Strong \mid \text{Play}=No) = 3/5$
  $P(\text{Play}=No) = 5/14$

- Decision making

$$P(Yes \mid \mathbf{x}') \approx [P(Sunny \mid Yes)P(Cool \mid Yes)P(High \mid Yes)P(Strong \mid Yes)]P(\text{Play}=Yes) = 0.0053$$
$$P(No \mid \mathbf{x}') \approx [P(Sunny \mid No)\, P(Cool \mid No)P(High \mid No)P(Strong \mid No)]P(\text{Play}=No) = 0.0206$$

Given the fact $P(Yes \mid \mathbf{x}') < P(No \mid \mathbf{x}')$, we label $\mathbf{x}'$ to be "*No*".

Naïve-BAYES for Continuous-valued Features

- Features are not always discrete, sometimes they are continuous
- Example: Temperature is a naturally continuous RV
  - Yes: 25.2, 19.3, 18.5, 21.7, 20.1, 24.3, 22.8, 23.1, 19.8
  - No: 27.3, 30.1, 17.4, 29.5, 15.1
- Problem: Very unlikely that we will find an exact value of temperature
  - E.g. P(Play = YES | Temp = 25.8), but Temp = 25.8 DOES NOT EXIST in data
- Solution: Model continuous probabilities with **Normal Distribution**

$$\hat{P}(X_j \mid C = c_i) = \frac{1}{\sqrt{2\pi}\sigma_{ji}} \exp\left(-\frac{(X_j - \mu_{ji})^2}{2\sigma_{ji}^2}\right)$$

μ_{ji} : mean (avearage) of feature values X_j of examples for which $C = c_i$

σ_{ji} : standard deviation of feature values X_j of examples for which $C = c_i$

Naïve-BAYES for Continuous-valued Features

$$\hat{P}(X_j \mid C = c_i) = \frac{1}{\sqrt{2\pi}\sigma_{ji}} \exp\left(-\frac{(X_j - \mu_{ji})^2}{2\sigma_{ji}^2}\right)$$

μ_{ji} : mean (avearage) of feature values X_j of examples for which $C = c_i$

σ_{ji} : standard deviation of feature values X_j of examples for which $C = c_i$

Learning Phase: for $\mathbf{X} = (X_1, \cdots, X_n)$, $C = c_1, \cdots, c_L$

Output: $n \times L$ normal distributions and $P(C = c_i)$ $i = 1, \cdots, L$

Test Phase: Given an unknown instance $\mathbf{X}' = (a_1', \cdots, a_n')$

- Instead of looking-up tables, calculate conditional probabilities with all the normal distributions achieved in the learning phrase

Naïve-BAYES for Continuous-valued Features

- Let's consider the temperature example again
  - Yes: 25.2, 19.3, 18.5, 21.7, 20.1, 24.3, 22.8, 23.1, 19.8
  - No: 27.3, 30.1, 17.4, 29.5, 15.1
- Estimate mean and variance for each class

$$\mu = \frac{1}{N}\sum_{n=1}^{N} x_n, \quad \sigma^2 = \frac{1}{N}\sum_{n=1}^{N}(x_n - \mu)^2$$

$$\mu_{Yes} = 21.64, \ \sigma_{Yes} = 2.35$$
$$\mu_{No} = 23.88, \ \sigma_{No} = 7.09$$

- **Learning Phase:** output two Gaussian models for P(temp|C)

$$\hat{P}(x \mid Yes) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x-21.64)^2}{2\times 2.35^2}\right) = \frac{1}{2.35\sqrt{2\pi}} \exp\left(-\frac{(x-21.64)^2}{11.09}\right)$$

$$\hat{P}(x \mid No) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x-23.88)^2}{2\times 7.09^2}\right) = \frac{1}{7.09\sqrt{2\pi}} \exp\left(-\frac{(x-23.88)^2}{50.25}\right)$$

Relevant Issues to consider for Naïve-BAYES

- **Violation of Independence Assumption**
  - For most real worlds tasks, P(X1, · · ·,Xn|C) ≠ P(X1|C) · · · P(Xn|C) [i.e. features are not independent]
  - Nonetheless, Naïve-BAYES works surprisingly well anyway
  - Check this out for a comparison against other models: notebook
- **Zero conditional probability Problem**
  - If no example contains the feature value $X_j = a_{jk}, \hat{P}(X_j = a_{jk} \mid C = c_i) = 0$
  - In this circumstance, $\hat{P}(x_1 \mid c_i) \cdots \hat{P}(a_{jk} \mid c_i) \cdots \hat{P}(x_n \mid c_i) = 0$ during test
  - For a remedy, conditional probabilities re-estimated with

$$\hat{P}(X_j = a_{jk} \mid C = c_i) = \frac{n_c + mp}{n + m}$$

n_c : number of training examples for which $X_j = a_{jk}$ and $C = c_i$

n : number of training examples for which $C = c_i$

p : prior estimate (usually, $p = 1/t$ for t possible values of X_j)

m : weight to prior (number of "virtual" examples, $m \geq 1$)

Continuous Naïve-BAYES Practice

Temperature	Play
25.2	Yes
27.3	No
19.3	Yes
18.5	Yes
30.1	No
21.7	Yes
17.4	No

Temperature	Play
20.1	Yes
24.3	Yes
29.5	No
22.8	Yes
23.1	Yes
19.8	Yes
15.1	No

Determine whether we will play or not on a day with temperature = 25.8, using continuous-valued Naive-Bayes.

$$\mu_{Yes} = 21.64,\ \sigma_{Yes} = 2.35$$
$$\mu_{No} = 23.88,\ \sigma_{No} = 7.09$$